

Mini Lesson Sample Script

Materials: linking cubes in two colors (about 20), personal white board

1. Use linking cubes to show that adding the same amount to both numbers keeps the difference the same.

Each linking cube is worth 10. Look at the two sticks. What is the difference? Now add one cube to each stick. What is the new difference?

Stick A: 7 cubes

Stick B: 4 cubes

Answer: $70 - 40 = 30$; $80 - 50 = 30$. The difference is the same, 30.

T: Here are two sticks of cubes. Each cube is worth 10. Let's count this one together by tens.

S: 10, 20, 30, 40, 50, 60, 70.

T: And this one?

S: 10, 20, 30, 40.

T: What is the difference between 70 and 40?
Give me a number sentence.

S: $70 - 40 = 30$.

T: Now watch. I'm going to add one cube to each stick. What's the new number sentence?

S: $80 - 50 = 30$.

T: What changed?

S: Both sticks got bigger by 10.

T: And what stayed the same?

	S: The difference. It's still 30. T: Right. When we add the same amount to both numbers, the difference does not change.
2. Use compensation with a tape diagram to subtract multiples of ten from three-digit numbers.	
Solve $340 - 180$ by adding the same amount to both numbers to make the problem easier. <i>Answer: Add 20 to each number. $360 - 200 = 160$, so $340 - 180 = 160$.</i>	T: Look at this tape diagram. The whole is 340 and one part is 180. We want to find the other part. How much would we need to add to 180 to get to a friendly hundred? S: 20. 180 plus 20 is 200. T: Good. So I'll add 20 to this bottom bar to make it 200. Now, if I only add to the bottom, did the difference stay the same? S: No. We have to add 20 to the top too. T: Why? S: Because the difference only stays the same if you add the same amount to both numbers. T: Yes. So 340 plus 20 is...? S: 360.

	T: Write the new equation on your board and solve it. S: $360 - 200 = 160$. T: So what is $340 - 180$? S: 160. It's the same, because the difference didn't change.
3. Use compensation to subtract a three-digit number close to a benchmark hundred.	
Solve $425 - 198$ by adding the same amount to both numbers to make the problem easier. <i>Answer: Add 2 to each number. $427 - 200 = 227$, so $425 - 198 = 227$.</i>	T: This time the number we're subtracting has some ones. What benchmark hundred is close to 198? S: 200. T: Try this one on your own. Write the new equation and solve. S: (Work.)